

Mid-Term Examination for Quantum Mechanism, Semester of 2025-2026(2)

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Problem 1

A particle of mass m has the wave function:

$$\Psi(x, t) = Ae^{(-a[\frac{mx^2}{\hbar} + it])}$$

where A and a are positive real constants.

- (a) Find A .
- (b) For what potential energy function $V(x)$ is this a solution to the Schrödinger equation?
- (c) Calculate the expectation values of x , x^2 , p and p^2 .
- (d) Find σ_x and σ_p . Is their product consistent with the uncertainty principle?

Problem 2

For a particle in the harmonic oscillator potential, its ground state is defined via the lowering operator

$$\hat{a}_- = \frac{1}{\sqrt{2\hbar m\omega}}(i\hat{p} + m\omega\hat{x})$$

by requiring $\hat{a}_- |0\rangle = 0$.

- (a) Find the normalized wave function of the ground state $\Psi_0(x) = \langle x | 0 \rangle$ in the position space.
- (b) For the first excited state $|1\rangle$, it can be obtained from $|1\rangle = \hat{a}_+ |0\rangle$, where the raising operator reads $\hat{a}_+ = \frac{1}{\sqrt{2\hbar m\omega}}(-i\hat{p} + m\omega\hat{x})$, find normalized $\langle x | 1 \rangle$.
- (c) Find the expectation values of the potential energy and the kinetic energy in the n -th stationary state.

“Hints: $\hat{a}_- |n\rangle = \sqrt{n} |n-1\rangle$ and $\hat{a}_+ |n\rangle = \sqrt{n+1} |n+1\rangle$.”

Problem 3

Orthonormal basis $|1\rangle, |2\rangle, |3\rangle$ with $\langle i | j \rangle = \delta_{ij}$.

$$|\alpha\rangle = i |1\rangle - 2 |2\rangle - i |3\rangle, \quad |\beta\rangle = i |1\rangle + 2 |3\rangle$$

- (a) Construct $\langle\alpha|$ and $\langle\beta|$.
- (b) Find inner products $\langle\alpha | \beta\rangle$ and $\langle\beta | \alpha\rangle$.
- (c) Find the matrix elements of $\hat{A} = |\alpha\rangle\langle\beta|$. Is it Hermitian?

Problem 4

A three-level quantum system has Hamiltonian and observables:

$$H = \hbar\omega \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad A = \lambda \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad B = \mu \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & 3 & 0 \end{bmatrix}$$

- (a) Find the eigenvalues and normalized eigenvectors of H , A , and B .

- (b) At $t = 0$, the system is in state $|S(0)\rangle = (c_1, c_2, c_3)^T$ with $|c_1|^2 + |c_2|^2 + |c_3|^2 = 1$. Find $\langle H \rangle$, $\langle A \rangle$, and $\langle B \rangle$ at $t = 0$.
- (c) Find $|S(t)\rangle$ at arbitrary time t . If energy is measured, what are the possible outcomes and their probabilities?

